

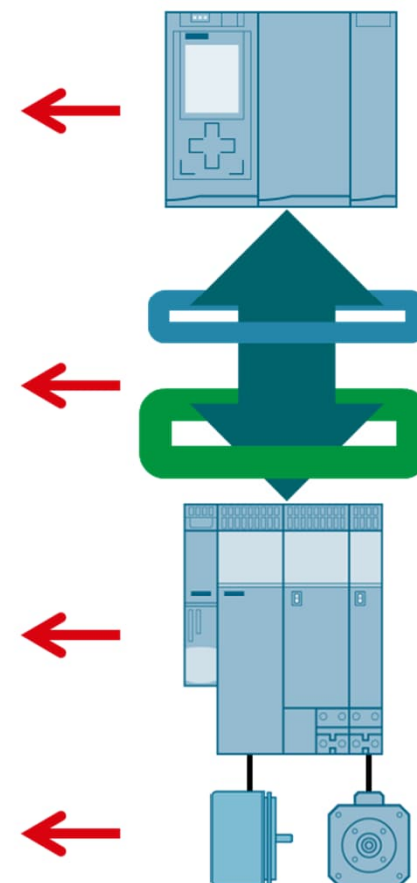
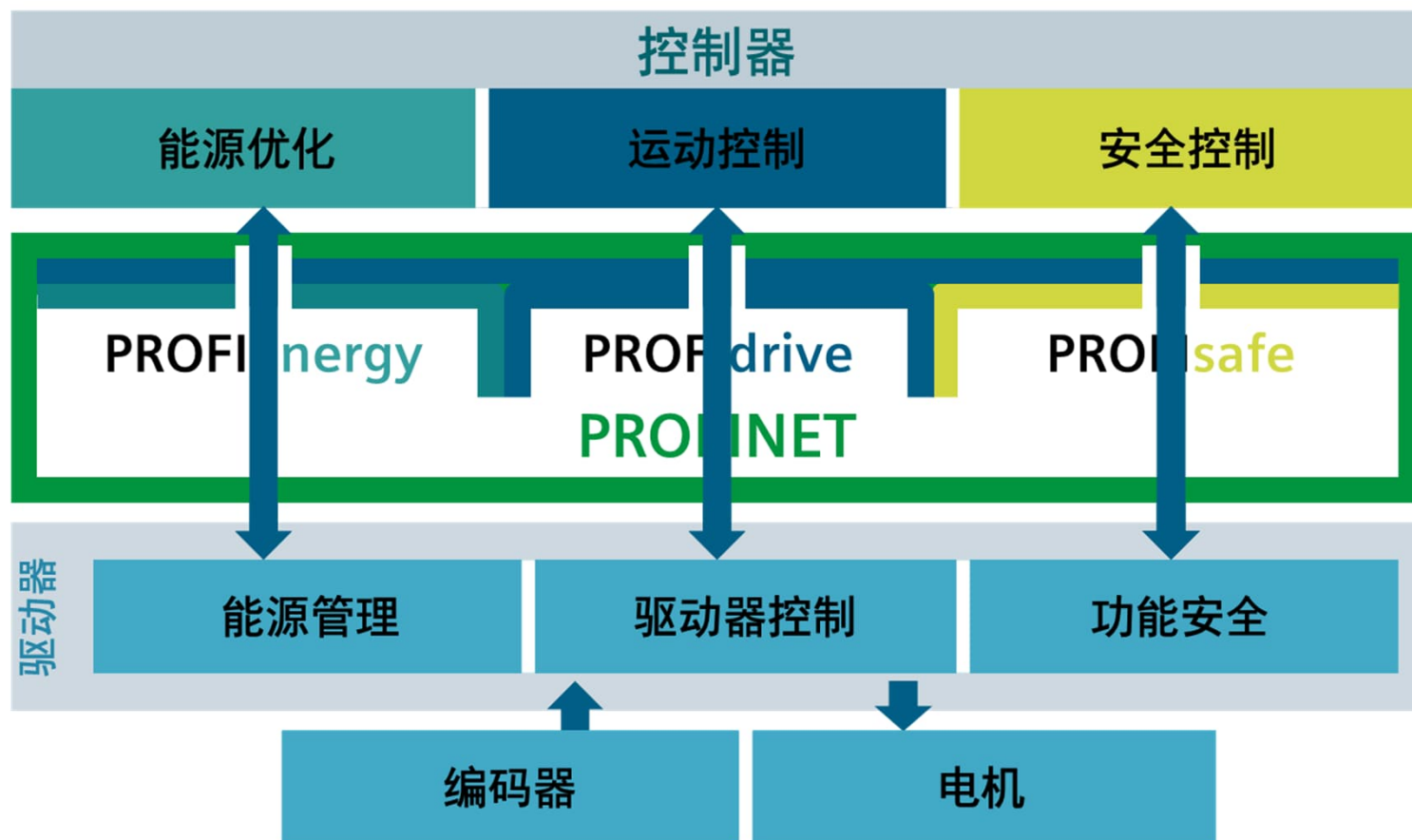
PROFIdrive运动控制行规开发经验分享

苏联
西门子数字化工业集团



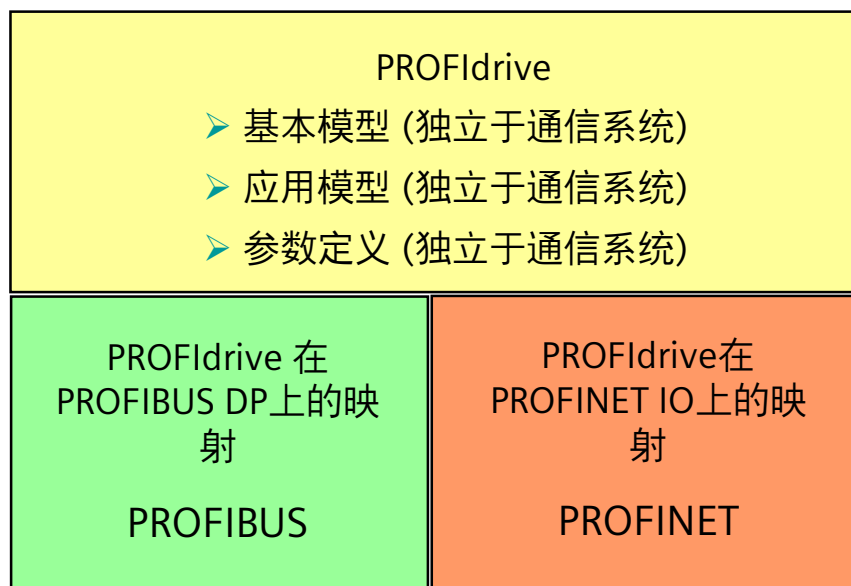
SIEMENS

什么是PROFIDrive？

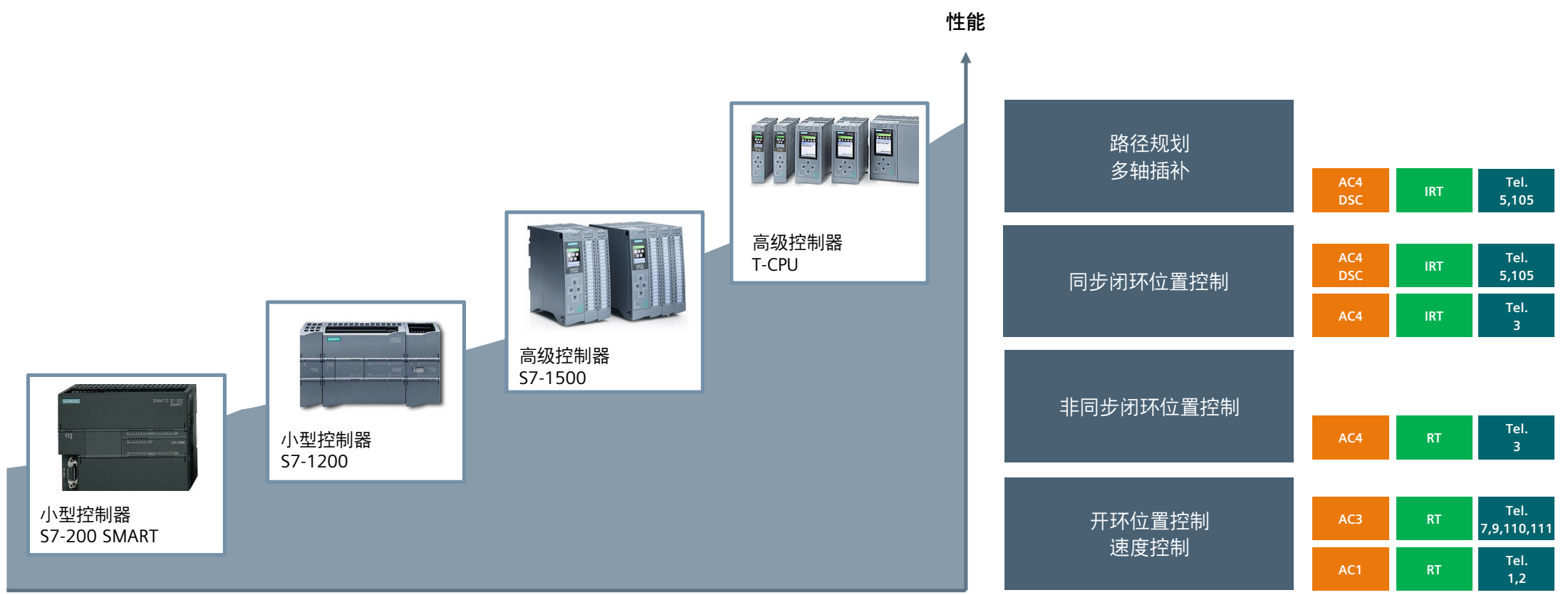


PROFIdrive行规优势

- ❑ PROFIdrive 是PROFINET 与PROFIBUS 两种通讯方式针对驱动技术应用的一种协议框架，也称作“行规”
- ❑ PROFIdrive 使得用户更快捷的实现各类驱动设备的控制
- ❑ PROFIdrive 的最大特点是互操作性 – 不同厂家的驱动器可以由同一个控制器通过总线进行控制



为什么开发支持PROFIDrive的驱动器？



支持PROFIDrive行规的驱动器能应用到什么场景？

应用类（AC）定义了运动控制领域的典型应用，PROFIdrive定义了6个应用类

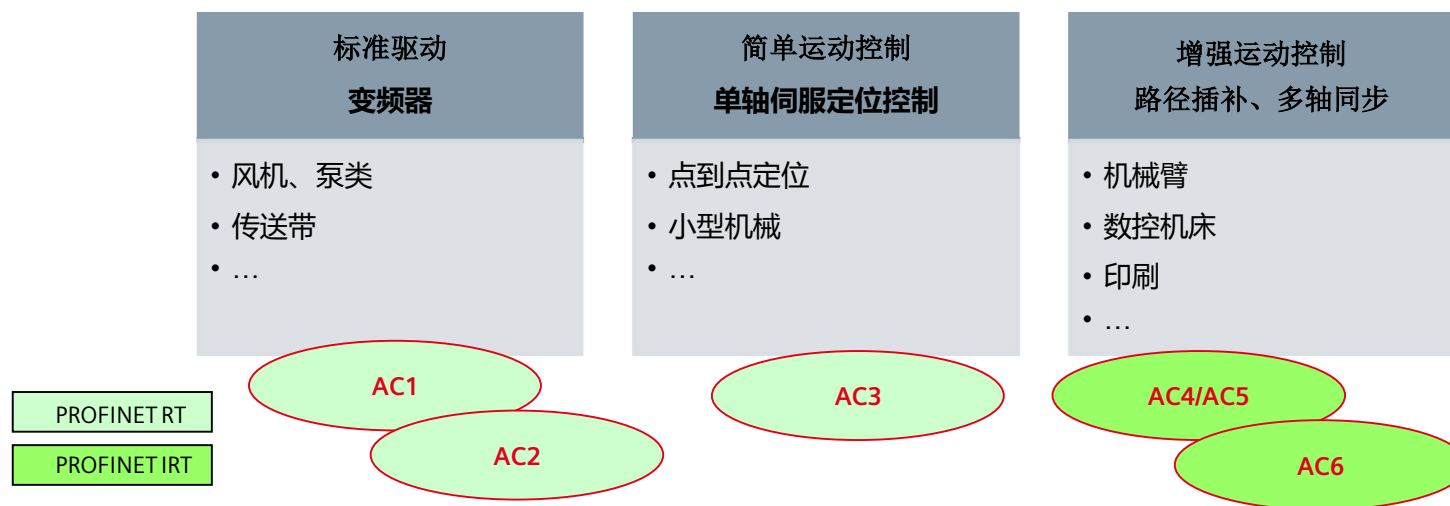
AC1：标准驱动

AC2：带分布式工艺控制器的标准驱动（连续过程）

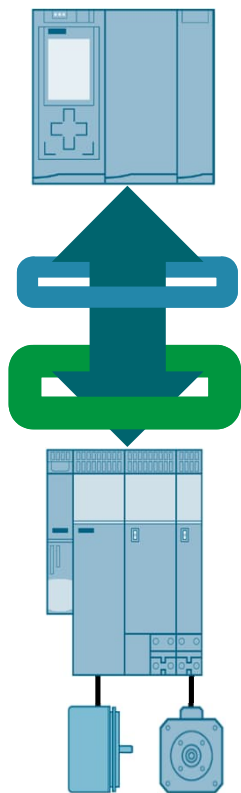
AC3：带有本地运动控制的单轴定位驱动

AC4/5：带有集中插补和速度设定值接口的多轴运动控制

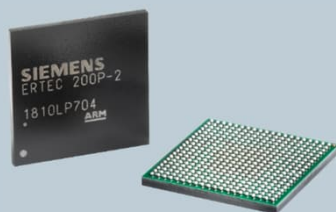
AC6：带有时钟处理或分布式同步的运动控制



如何快速开发支持PROFIDrive行规的驱动器硬件？



• 硬件



ERTEC 200P-2

- 集成了支持IRT的2端口交换机
- 符合C类（性能提升）
- 高负载-用户程序可在集成的ARM中运行
- 循环时间可达125 μs



基于ERTEC芯片的参考设计板

- 设计文件免费提供
- 低价格，精简的硬件设计
- 尺寸小巧，便于嵌入到已有产品中
- 本地化的支持团队



PN Core 快速开发模块

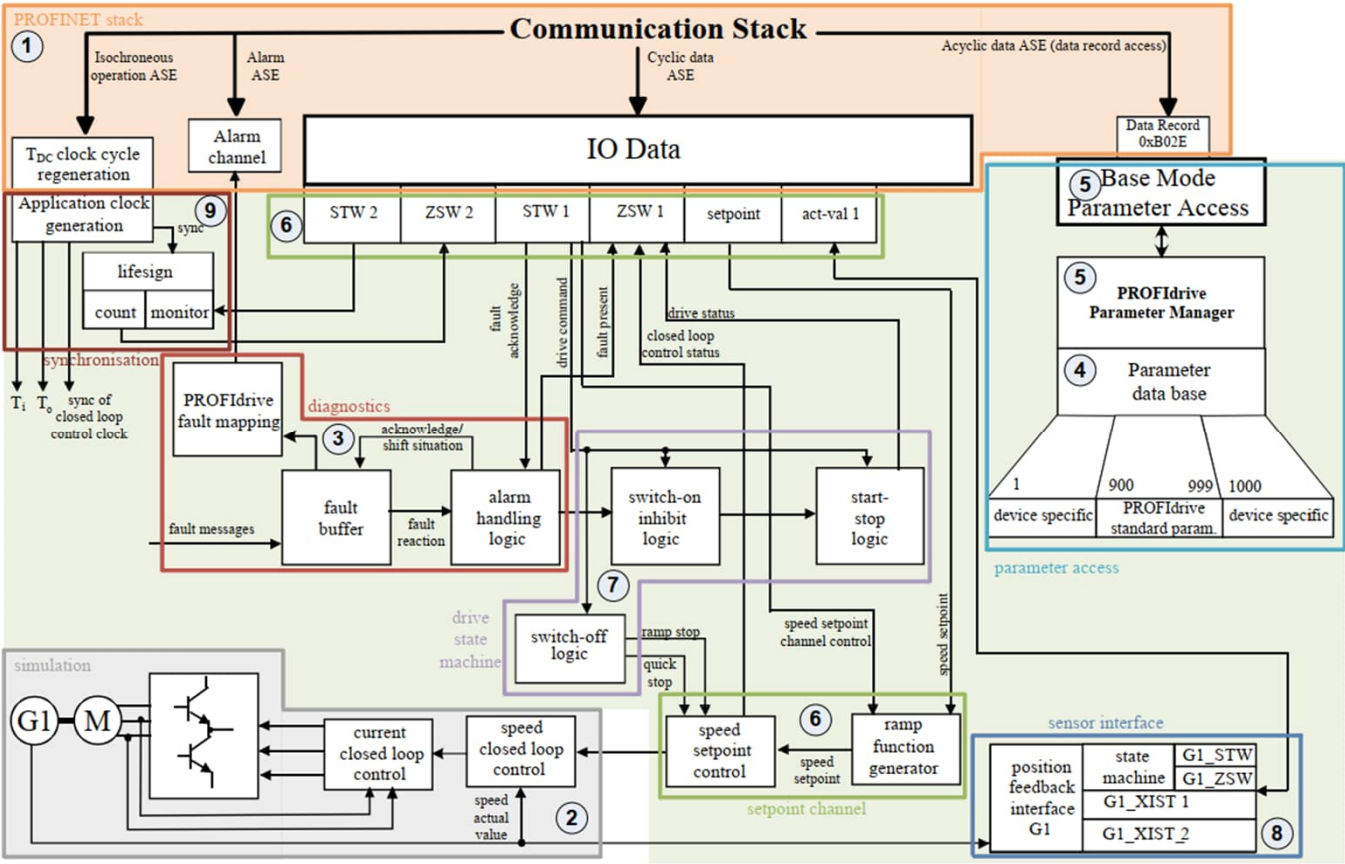
- 灵活-接口全部引出
- 高性能-提供高速主机接口参考方案
- 易于订制-外置以太网和调试接口
- 提供底板的软硬件设计资料

• 软件



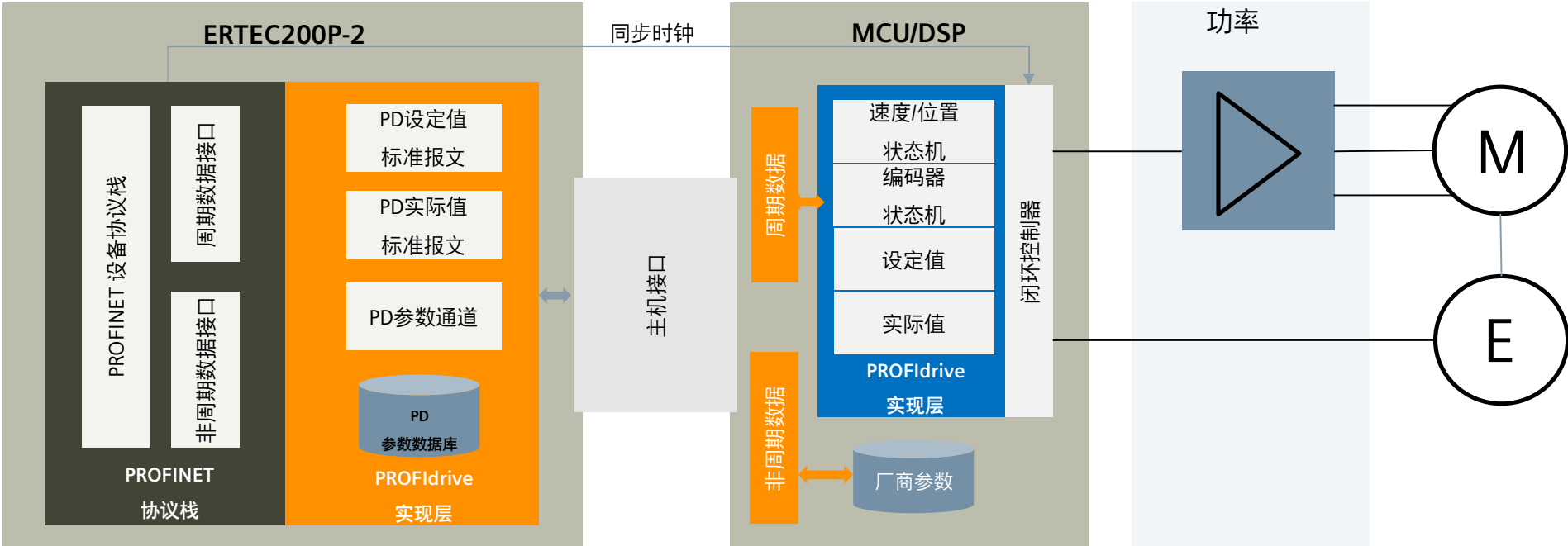
- 源代码提供PROFINET完整的协议栈
- ERTEC 200P-2的样例程序
- 免授权的开发工具和实时的嵌入式可配置实时操作系统
- 免费的网络分析工具

PROFIDrive行规软件框架？



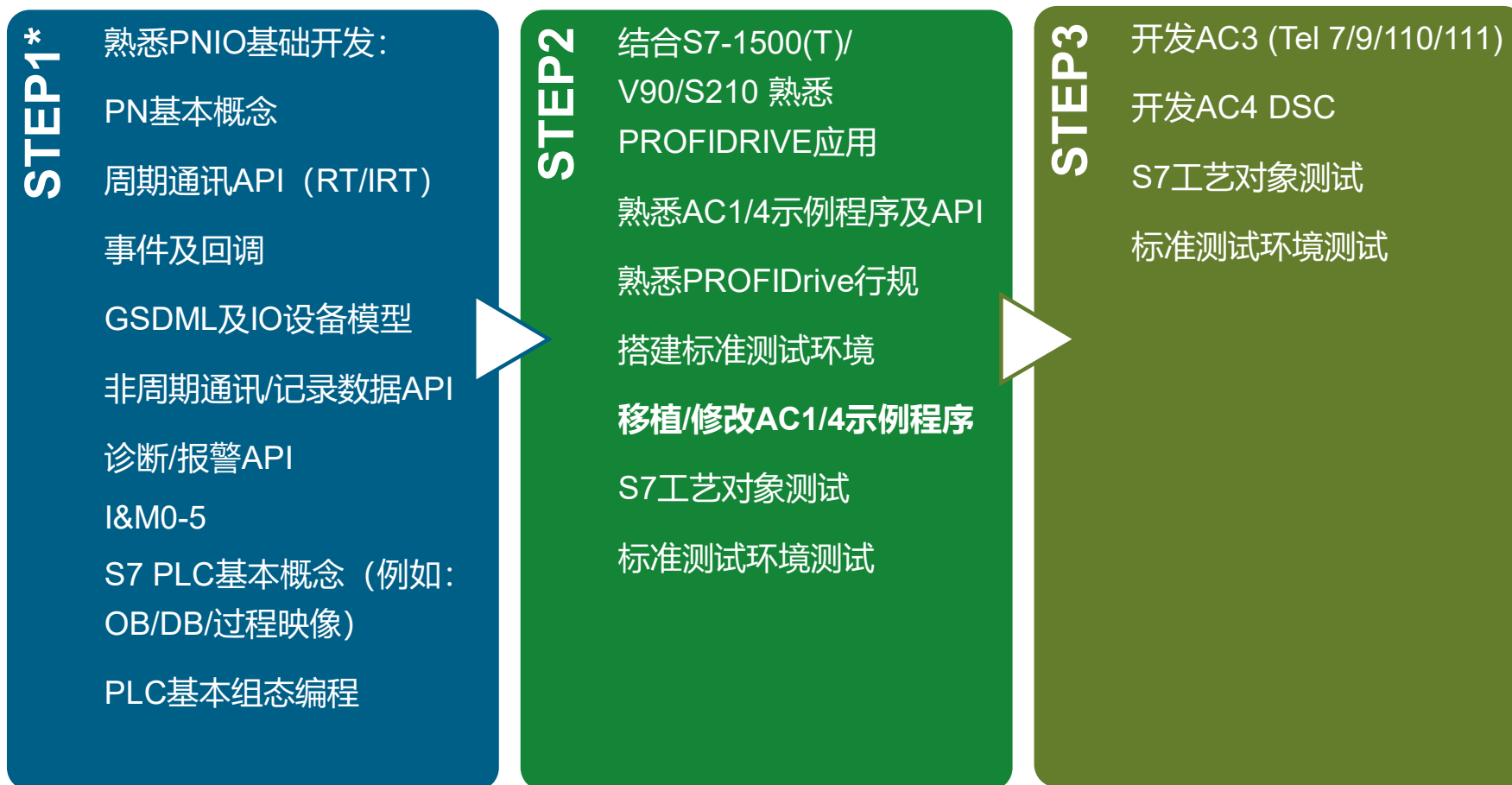
File	Brief description	Fig. No.
usrnod_main_pdrvac4.c	Main program of the PROFINET example Start of the PROFINET IO stack, initialization of the PROFIDrive application, login of the call-up of the PROFIDrive application at the isochronous interrupt , main loop with start of functions via keyboard of PROFINET functions (for example, firmware updates)	①
iodapi_event_pdrvac4.c	PROFINET interface with messages from PROFINET events to the application Contains functions which the PROFINET IO stack calls when events such as connection/disconnection, alarm reception, etc. occur and thus informs the application.	①
PnUtr_Api_pdrvac4.c	PROFINET subroutines for the user example. If necessary, the functions contained can be used as a function library in the customer application.	①
pdrv_application_ac4.c pdrv_application_ac4.h	PROFIDrive application 1ms time slice with call of the other PROFIDrive modules, simulation of the control system, terminal outputs for PROFIDrive events, implementation of the application parameters PNU100-PNU819	②
pdrv_diagnostics_ac4.c pdrv_diagnostics_ac4.h	PROFIDrive diagnostics with fault and warning handling	③
pdrv_parameter_ac4.inc	Macro file with list of the implemented PROFIDrive parameters	④
pdrv_pardatabase_ac4.c pdrv_pardatabase_ac4.h	PROFIDrive parameter database Creates the parameter database from pdrv_parameter.inc	④
pdrv_parmanager_ac4.c pdrv_parmanager_ac4.h	PROFIDrive Parameter Manager Realizes the PROFIDrive Base Mode Parameter Access	⑤
pdrv_setpointchannel_ac4.c pdrv_setpointchannel_ac4.h	PROFIDrive setpoint channel Implements access to the signals of standard frames 1 & 2; selection of the setpoint source based on the general state, ramp generator	⑥
pdrv_statemachine_ac4.c pdrv_statemachine_ac4.h	PROFIDrive General State Machine	⑦
pdrv_sensor_ac4.c pdrv_sensor_ac4.h	PROFIDrive encoder: Implements the encoder state machine and the simulated absolute value encoder.	⑧
pdrv_synchronization_ac4.c pdrv_synchronization_ac4.h	PROFIDrive synchronization Implements the synchronization via signs of life with the evaluation of CACF	⑨
pdrv_types_ac4.h	Definitions of PROFIDrive data types	

如何快速开发支持PROFIDrive行规的驱动器软件？



行规源代码、文档及示例程序免费提供！			
PROFIdrive 通用		PROFIdrive (AC 1)	PROFIdrive (AC 3)
状态机	诊断通道	速度设定值	位置控制(PROG/MDI)
参数通道		速度状态机	位置状态机
		诊断通道	诊断通道
		参数通道	参数通道
		PROFIdrive (AC 4)	
		编码器通道	
		速度设定值	
		同步	
		速度状态机	
		诊断通道	
		参数通道	

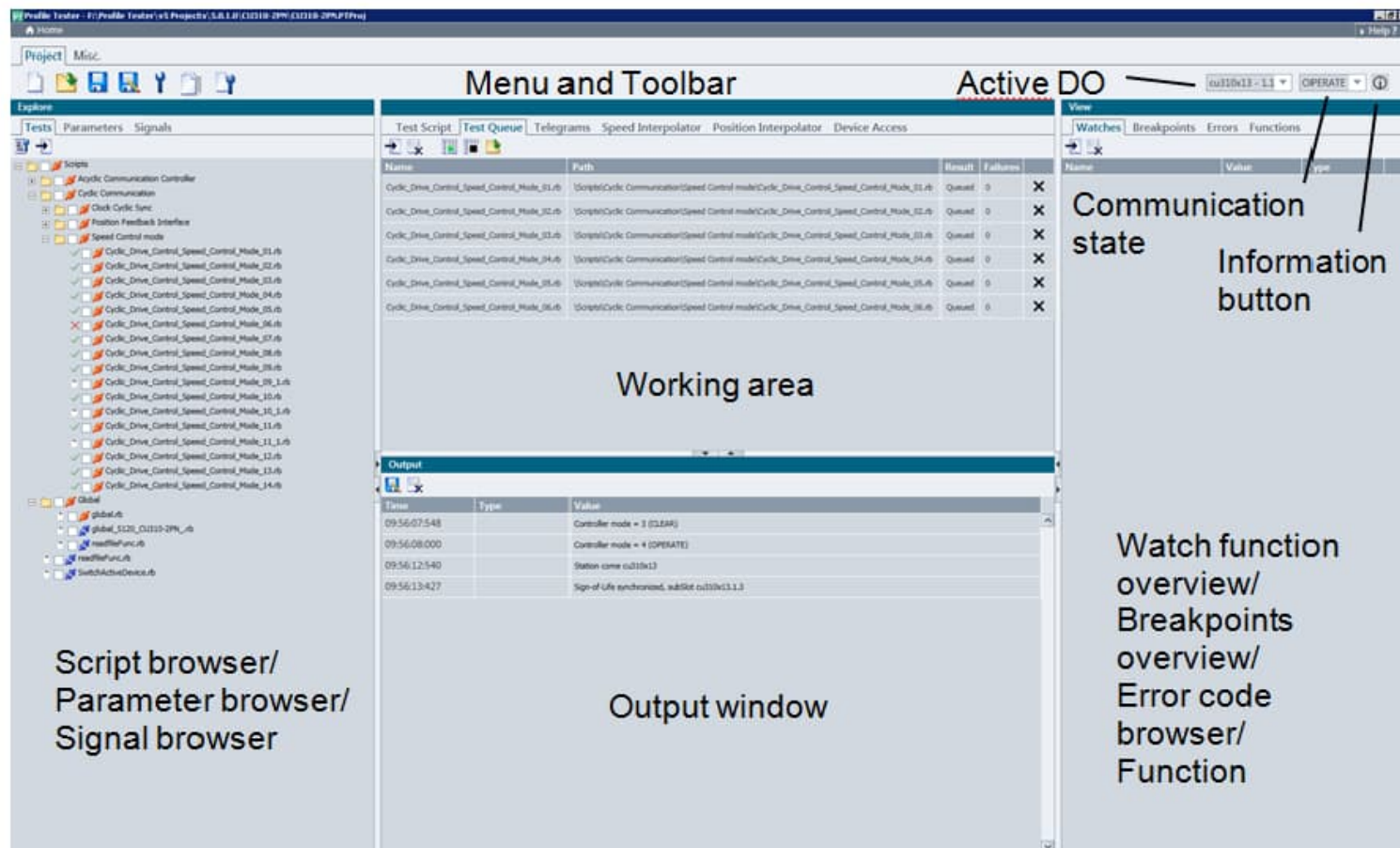
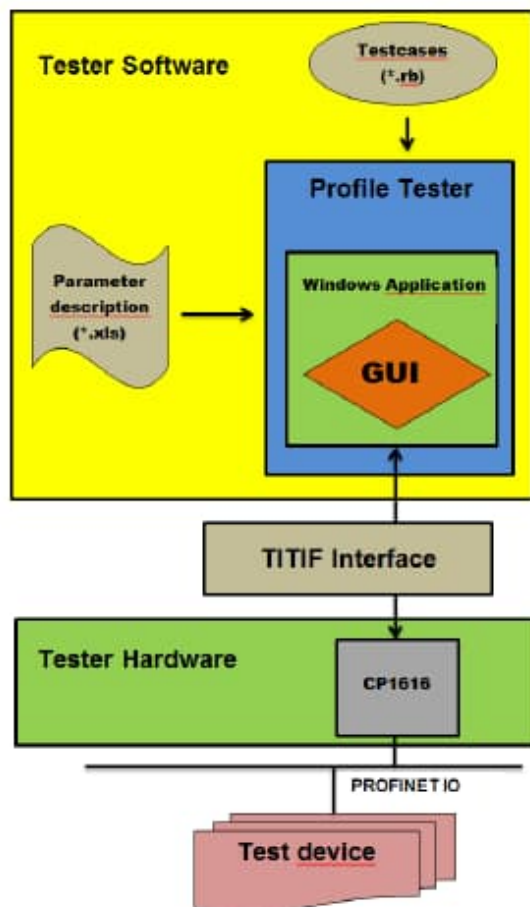
建议的开发流程



必备软件：TIA Portal, Proneta, PROFINET GSD Checker, PD Profile Tester

*:STEP1 可参考公众号文章

产品上市之前需要做什么？



谢谢！



SIEMENS

SIEMENS

午餐



酒店一层蓉咖啡厅

下午会议开始时间14:00